

Tutorial-1 ✓

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A smart farming system predicts whether a plant will grow well (1) or poorly (0) based on daily sunlight hours and water amount in liters.

Dataset:

sunlight Hours (x_1)	water liters (x_2)	Bias (x_3)	Target (Good Growth = 1 / Poor = 0)
3	1	1	(1) 0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1

① Understand and Explain the data

Inputs (x_1 & x_2) output (Target)

mean of $x_1 = \mu_{x_1} = 6.6$

mean of $x_2 = \mu_{x_2} = 3$

* Variance; $\sigma^2 = \frac{1}{N} \sum (x_i - \mu)^2$

where $N = \text{No. of Data}$

$x_i = \text{Input}$

$\mu = \text{mean}$

$$\frac{(x_i - \mu_{x_1})^2 + (x_i - \mu_{x_2})^2 + \dots}{\text{no. of } x \text{ values}}$$

$$\begin{aligned}
 &= \frac{(x_{11} - \mu x_1)^2 + (x_{12} - \mu x_1)^2 + (x_{13} - \mu x_1)^2 + (x_{14} - \mu x_1)^2 + (x_{15} - \mu x_1)^2}{5} \\
 &= \frac{(3-6.6)^2 + (8-6.6)^2 + (5-6.6)^2 + (10-6.6)^2 + (7-6.6)^2}{5} \\
 &= 5.84
 \end{aligned}$$

$$\Rightarrow \frac{(x_i - \mu_i)^2 + (x_i - \mu_{x_i})^2 + \dots}{\text{no. of values}}$$

$$= \frac{(x_{21} - \mu x_2)^2 + (x_{22} - \mu x_2)^2 + (x_{23} - \mu x_2)^2 + (x_{24} - \mu x_2)^2 + (x_{25} - \mu x_2)^2}{5}$$

$$= \frac{(1-3)^2 + (4-3)^2 + (2-3)^2 + (5-3)^2 + (3-3)^2}{5}$$

$$= \frac{(-2)^2 + (1)^2 + (-1)^2 + (2)^2 + (0)^2}{5}$$

$$= \frac{4+1+1+4+0}{5} = \frac{10}{5} = 2$$

variance of $x_1 \Rightarrow \sigma^2 x_1 = 5.84$

variance of $x_2 \Rightarrow \sigma^2 x_2 = 2$

$$\sigma^2 x_1 > \sigma^2 x_2$$

standard deviation of

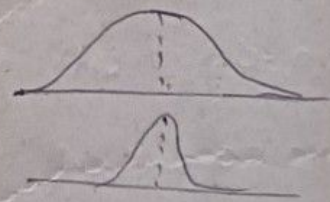
$$\sqrt{\sigma x_1^2} = \sigma x_1 = 2.40$$

standard deviation of

$$\sqrt{\sigma x_2^2} = \sigma x_2 = 1.414$$

The factor whose variance is higher is important

$\therefore$ The Sunlight > water
Hours Liters



ML model

perceptron model:

It is a simple Neural Network model

It is a simple decision making model that can Answer Yes/No

perceptron:

① Inputs: x_1, x_2

② weights associated with input: w_1, w_2

③ Bias $\rightarrow$ It gives the model flexibility (b)
almost it is 1

④ weighted sum: $z = w_1 x_1 + w_2 x_2 + b = \sum_i w_i x_i + b$
 $w_1, w_2 = 0$

⑤ Activation function $\rightarrow$ It is an mathematical (step function) function followed by ml

step function is a easiest function with (0,1)

Activation function: $y^{\wedge} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$

⑥ Updated weight:

$$w_i^{\text{new}} = w_i^{\text{old}} + \eta (y - y^{\wedge}) x_i$$

η (eta)

$$z = 0(3) + 0(1) + 1$$

$$z = 1 < 0 \quad 0$$

$$0.5 \quad 0.5 \quad 1$$

y = Actual Target/output

$$= 0 + 1(0-1) \quad 3$$

$y^{\wedge}$ = predicted target/output

$$= 0 \neq 3$$

$$w_1 = -3$$

η = Learning Rate

$$w_2 = 0 + 1(0-1) \quad 1$$

$$w_2 = -1$$

prediction:

① Initial

$$z = (-3)(3) + (-1)(1) + 1$$

Let; $w_1 = 0, w_2 = 0, \eta = 1$ we can take 0-9 also.

$$-22 - 1 + 1 < 0 \quad 0$$

Inputs = $x_1 = 3, x_2 = 1, b = 1$

$$z = ?$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$(0)(3) + (0)(1) + 1 = 1$$

$$z = 1, \text{ so check is } y^{\wedge} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$$

$$y^{\wedge} = 1$$

but the target is "a" so we want to change the weights.

$$w_1^{\text{new}} = 0 + 1(0 - 1) \cdot 3 = -3$$

$$w_2^{\text{new}} = 0 + 1(0 - 1) \cdot 1 = -1$$

2nd case for 2nd sunlight & waves inputs

Inputs: $x_1 = 8, x_2 = 4, b = 1, w_1 = -3, w_2 = -1, \eta = 1, b = 1$

$$z = (-3)(8) + (-1)(4) + 1$$

$$= -24 - 4 + 1$$

$$= -28 + 1$$

$$= -27$$

check $y^{\wedge}$

$$\boxed{z = -27}, \quad \boxed{y^{\wedge} = 0}$$

But in Dataset of 2nd row the target is 1 so it do not matched.

$$w_1^{\text{new}} = (-3) + 1(1 - 0)8$$

$$= -3 + (1 - 0)8$$

$$= -3 + 8$$

$$w_1^{\text{new}} = 5$$

$$w_2^{\text{new}} = -1 + 1(1 - 0)(4)$$

$$w_2^{\text{new}} = -1 + 4$$

$$w_2^{\text{new}} = 3$$

$$x_1 = 5 \quad x_2 = 2 \quad b = 1 \quad w_1 = 5 \quad w_2 = 3 \quad \eta = 1$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$(5)(5) + (3)(2) + 1$$

$$25 + 6 + 1$$

$$z = 32 \quad z > 0 \Rightarrow y^n = 1$$

$$w_{1\text{new}} = (5) + 1(0-1)(5)$$

$$w_{1\text{new}} = 5 + (-5)$$

$$\boxed{w_{1\text{new}} = 0}$$

$$w_{2\text{new}} = (3) + 1(0-1)(2)$$

$$w_{2\text{new}} = 3 - 2$$

$$\boxed{w_{2\text{new}} = 1}$$

$$x_1 = 10, \quad x_2 = 5, \quad b = 1, \quad w_1 = 0, \quad w_2 = 1, \quad z = 1$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$= (0)(10) + (1)(5) + 1$$

$$= 0 + 5 + 1$$

$$\boxed{z = 6}$$

$$z > 0 \text{ so } \boxed{y^n = 1}$$

$\therefore$ activation function = old target